

## C5.2 How do chemists find the composition of unknown samples? (*separate science only*)

| Teaching and learning narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Assessable learning outcomes<br><i>Learners will be required to:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <p>Chemists use qualitative analysis to identify components in a sample. The procedures have a wide range of applications, including testing chemicals during manufacturing process, testing mineral samples, checking for toxins in waste, environmental testing of water, testing soils.</p> <p>Chemists use sampling techniques to make sure that the samples used for the analysis are representative and will identify any variations in the bulk of the material that is represented in the analysis (1aS1).</p> <p>Laboratory analysis can be used to identify the metal cations and the anions in salts. Cations can be identified using flame tests or by adding dilute sodium hydroxide. Anions can be identified using a range of dilute reagents.</p> <p>Instrumental analysis is widely used in research and in industry. Emission spectroscopy is a technique which relies on looking at the spectrum of light emitted from a hot sample. Each element gives a unique pattern of lines. Elements can be identified by matching the patterns and wavelengths of lines to reference data from known elements. Emission spectroscopy is used to identify elements in stars and in substances such as steel in industry.</p> <p>Instrumental analysis is preferred due to its greater sensitivity, speed and accuracy. Data is automatically recorded. However, the technology is expensive and is not as freely available as the standard glassware used in laboratory analysis.</p> | 1. describe the purpose of representative sampling in qualitative analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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### Linked learning opportunities

#### Specification links:

- Symbols, formulae and ionic equations. (C2.4 and C3.2)

#### Ideas about Science:

- Suggest equipment and techniques and a strategy to carry out qualitative analysis (1aS1)